

Session2-3.R

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```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
## Set Working Directory
setwd("~/Desktop/MLWBA")
## Import Dataset
insurance = read.csv("insurance_LR1.csv")
summary(insurance)
```

```
##      age      sex      bmi      children
##  Min.    :18.00  Length:1338  Min.    :15.96  Min.    :0.000
##  1st Qu.:27.00  Class  :character  1st Qu.:26.27  1st Qu.:0.000
##  Median :39.00  Mode   :character  Median :30.40  Median :1.000
##  Mean    :39.21                Mean    :30.66  Mean    :1.095
##  3rd Qu.:51.00                3rd Qu.:34.70  3rd Qu.:2.000
##  Max.    :64.00                Max.    :53.13  Max.    :5.000
##                                NA's    :2
##      smoker      region      charges
##  Length:1338    Length:1338    Min.    : 1122
##  Class  :character  Class  :character  1st Qu.: 4740
##  Mode   :character  Mode   :character  Median   : 9382
##                                Mean    :13270
##                                3rd Qu.:16640
##                                Max.    :63770
##
```

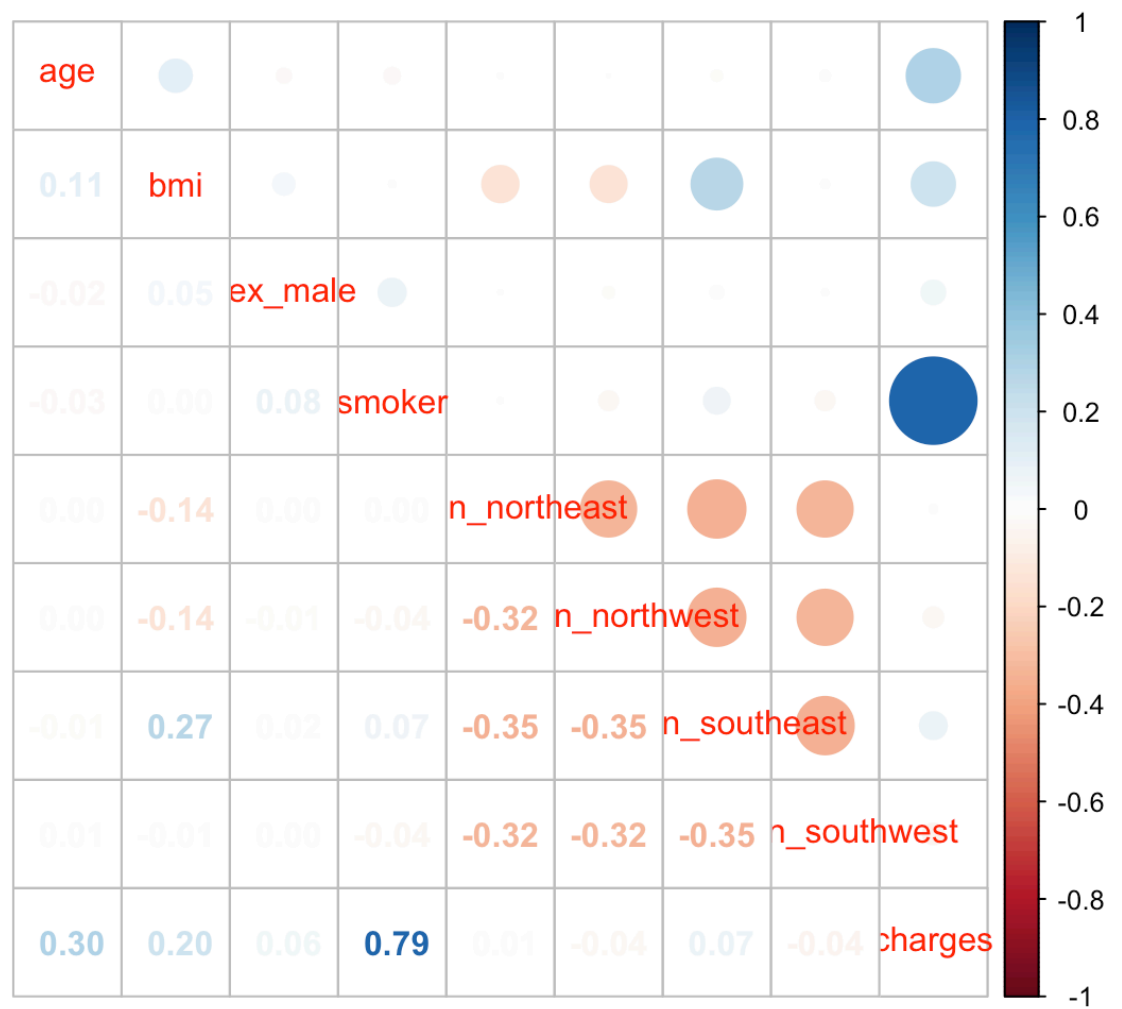
```
View(insurance)
insurance = read.csv("insurance_LR1.csv",stringsAsFactors = TRUE)
View(insurance)
summary(insurance)
```

```
##          age          sex          bmi          children          smoker
##  Min.      :18.00    female:662    Min.      :15.96    Min.      :0.000    no :1064
##  1st Qu.:27.00    male  :676    1st Qu.:26.27    1st Qu.:0.000    yes: 274
##  Median :39.00                                Median :30.40    Median :1.000
##  Mean    :39.21                                Mean    :30.66    Mean    :1.095
##  3rd Qu.:51.00                                3rd Qu.:34.70    3rd Qu.:2.000
##  Max.    :64.00                                Max.    :53.13    Max.    :5.000
##
##                                NA's      :2
##          region          charges
##  northeast:324    Min.      : 1122
##  northwest:325    1st Qu.: 4740
##  southeast:364    Median   : 9382
##  southwest:325    Mean      :13270
##                                3rd Qu.:16640
##                                Max.      :63770
##
```

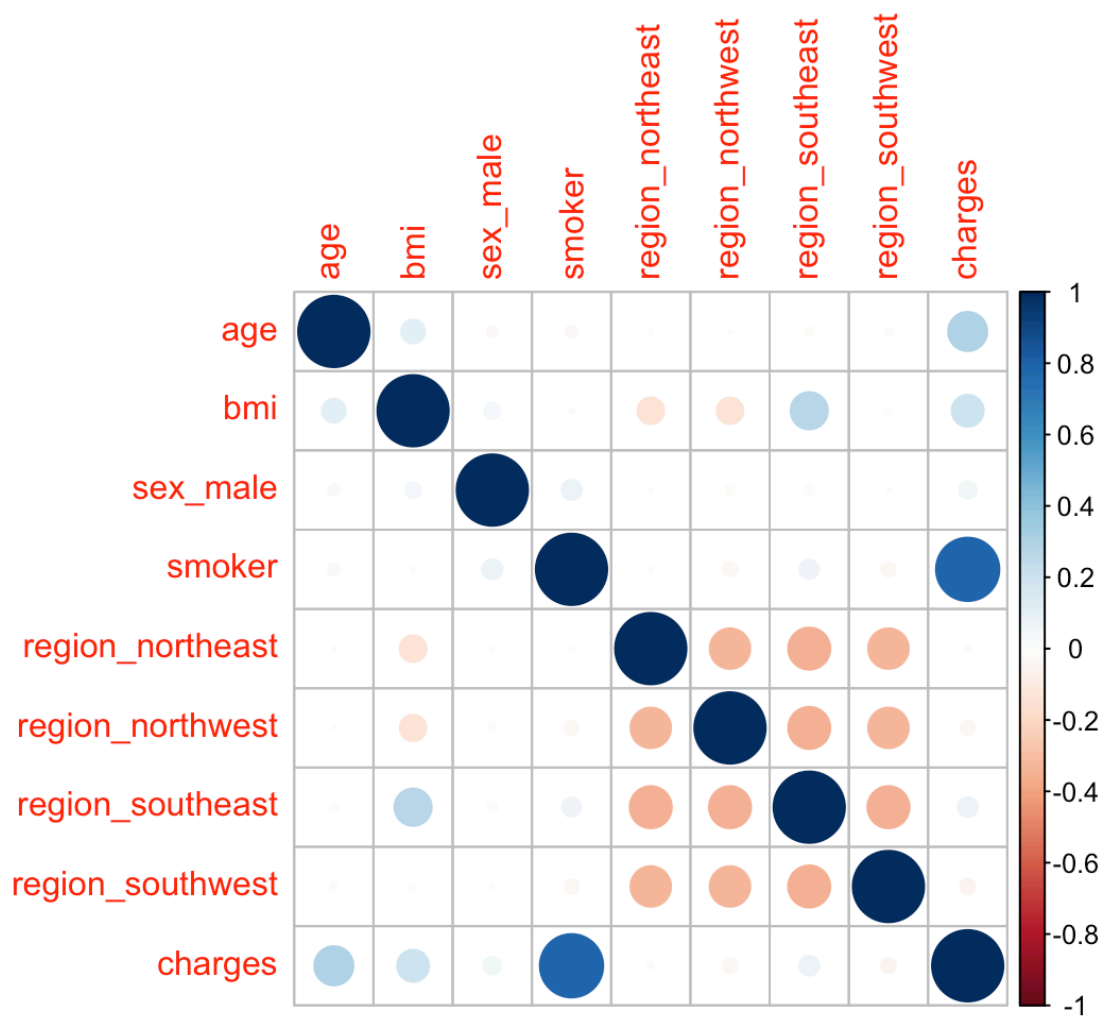
```
## convert into Categorical Data
insurance$children = as.factor(insurance$children)
insurance = na.omit(insurance)
## Step 2: Create Dummy Variable
## install.packages("fastDummies")
library(fastDummies)
insurance = dummy_cols(insurance, select_columns = c ("sex", "region"))
## label encoding
insurance$smoker = ifelse (insurance$smoker=="yes",1,0)
## Step 3: Correlation
cr = cor(insurance[c("age","bmi","sex_male","smoker","region_northeast","region_northwest",
                    "region_southeast","region_southwest","charges")])
cr
```

```
##              age              bmi      sex_male      smoker
## age          1.0000000000  0.109241926 -0.020534700 -0.025199587
## bmi          0.1092419260  1.0000000000  0.046456676  0.003724879
## sex_male     -0.0205347004  0.046456676  1.000000000  0.077007886
## smoker       -0.0251995874  0.003724879  0.077007886  1.000000000
## region_northeast 0.0022825014 -0.138219181 -0.001589578  0.002382745
## region_northwest -0.0006998183 -0.136261094 -0.012068244 -0.036544052
## region_southeast -0.0110021671  0.270477280  0.016388733  0.068979499
## region_southwest  0.0098259531 -0.006236044 -0.003348203 -0.037391955
## charges        0.2988485844  0.198349821  0.058226416  0.787176783
##              region_northeast region_northwest region_southeast
## age          0.002282501      -0.0006998183      -0.01100217
## bmi          -0.138219181      -0.1362610936      0.27047728
## sex_male     -0.001589578      -0.0120682444      0.01638873
## smoker       0.002382745      -0.0365440519      0.06897950
## region_northeast 1.000000000      -0.3201581028      -0.34560430
## region_northwest -0.320158103      1.0000000000      -0.34560430
## region_southeast -0.345604303      -0.3456043031      1.000000000
## region_southwest -0.320810336      -0.3208103356      -0.34630838
## charges        0.005851891      -0.0397312228      0.07482464
##              region_southwest      charges
## age          0.009825953  0.298848584
## bmi          -0.006236044  0.198349821
## sex_male     -0.003348203  0.058226416
## smoker       -0.037391955  0.787176783
## region_northeast -0.320810336  0.005851891
## region_northwest -0.320810336 -0.039731223
## region_southeast -0.346308375  0.074824639
## region_southwest  1.000000000 -0.043733841
## charges        -0.043733841  1.000000000
```

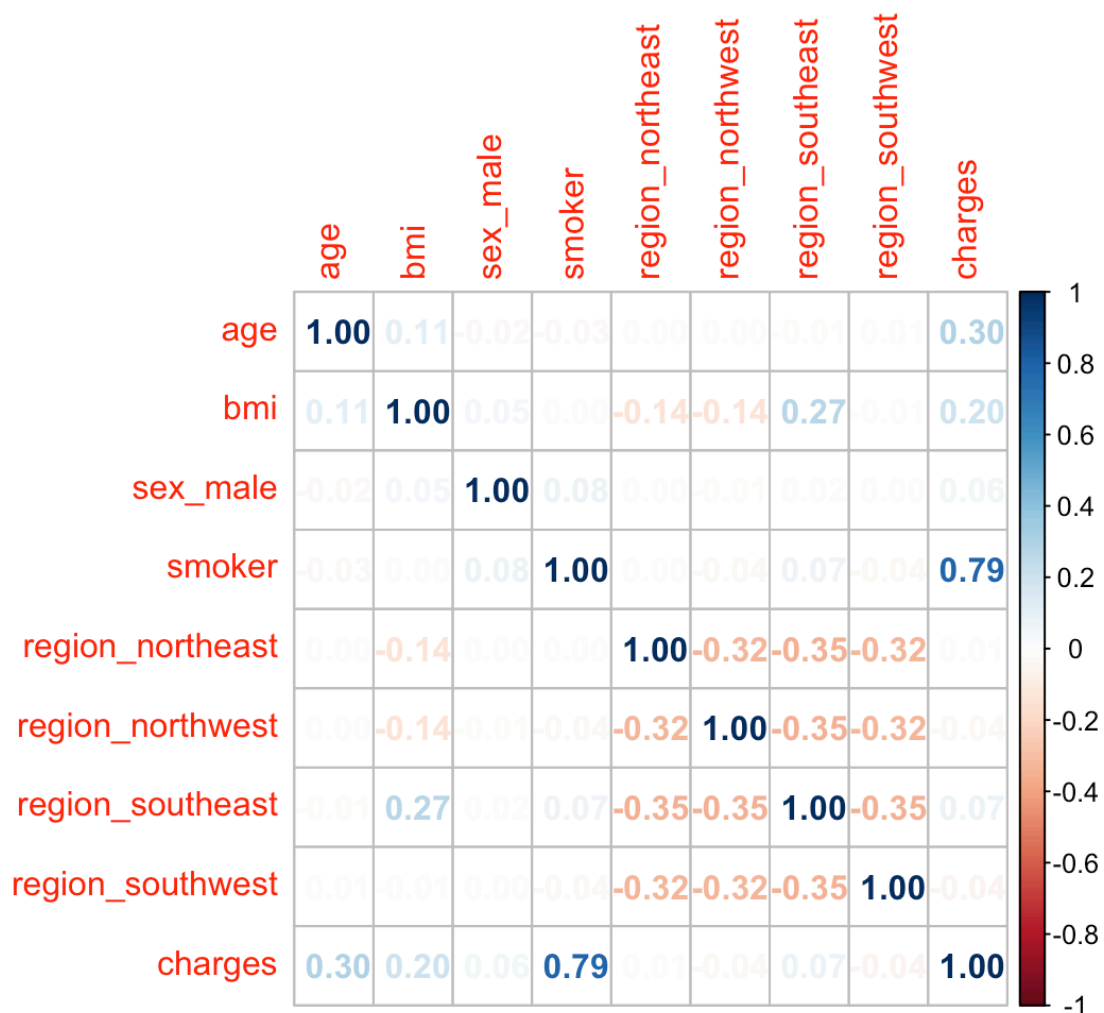
```
corrplot.mixed(cr)
```



```
corrplot(cr,type = "full")
```



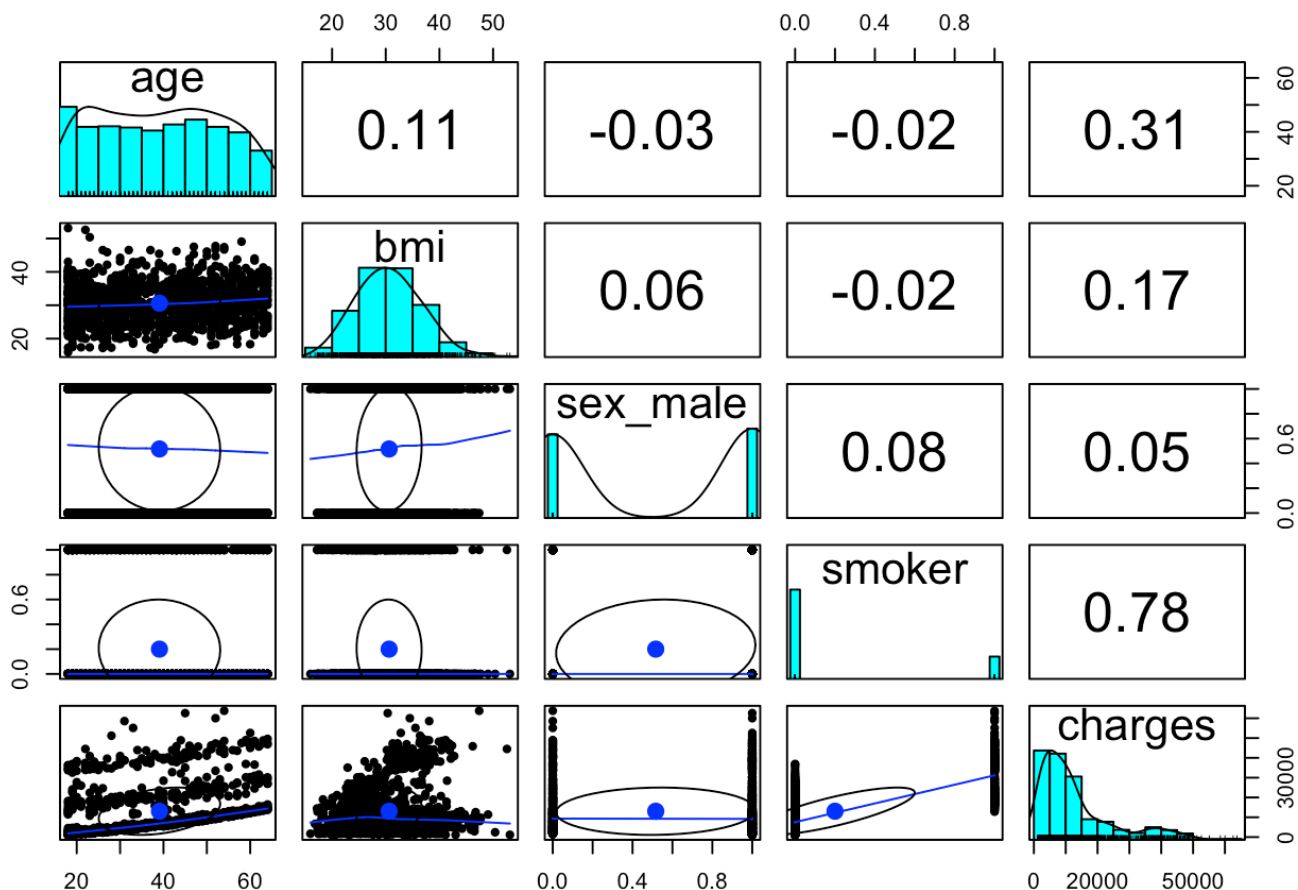
```
corrplot(cr,method = "number")
```



```
## help
?corrplot
## Step 4: Training & Testing (to do prediction)
## install.packages("caTools")
library(caTools)
set.seed(111)
split = sample.split(insurance$charges, SplitRatio = 0.85)
training_data = subset(insurance,split == "TRUE")
testing_data = subset(insurance,split == "FALSE")
View(training_data)
View(testing_data)

## Note ##

## DV has to be normally distributed -> Hetero scarcity
## DV and IV must have correlation -> Insignificant Variable
## Two IV shouldn't be highly correlated -> Multicollinearity
library(psych)
pairs.panels(training_data[c("age","bmi","sex_male","smoker","charges")])
```



```
## Step 5:building the model on training dataset
modell = lm(charges ~ age, data = training_data)
summary(modell)
```

```
##
## Call:
## lm(formula = charges ~ age, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7884  -6481  -5775   5344  47969
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2704.15    1000.28   2.703  0.00697 **
## age          264.89      24.09  10.994 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11380 on 1133 degrees of freedom
## Multiple R-squared:  0.09639,    Adjusted R-squared:  0.0956
## F-statistic: 120.9 on 1 and 1133 DF,  p-value: < 2.2e-16
```

```
model2 = lm(charges ~ age+bmi, data = training_data)
summary(model2)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12920  -6811  -5139   6559   48121
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5397.18     1877.57  -2.875  0.00412 **
## age           251.68       23.98  10.497 < 2e-16 ***
## bmi           281.40       55.42   5.077 4.48e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11250 on 1132 degrees of freedom
## Multiple R-squared:  0.1165, Adjusted R-squared:  0.115
## F-statistic: 74.64 on 2 and 1132 DF,  p-value: < 2.2e-16
```

```
## vif = 1/(1-R^2) {if vif > 5 -> Multicollinearity}
library(car)
```

```
## Loading required package: carData
```

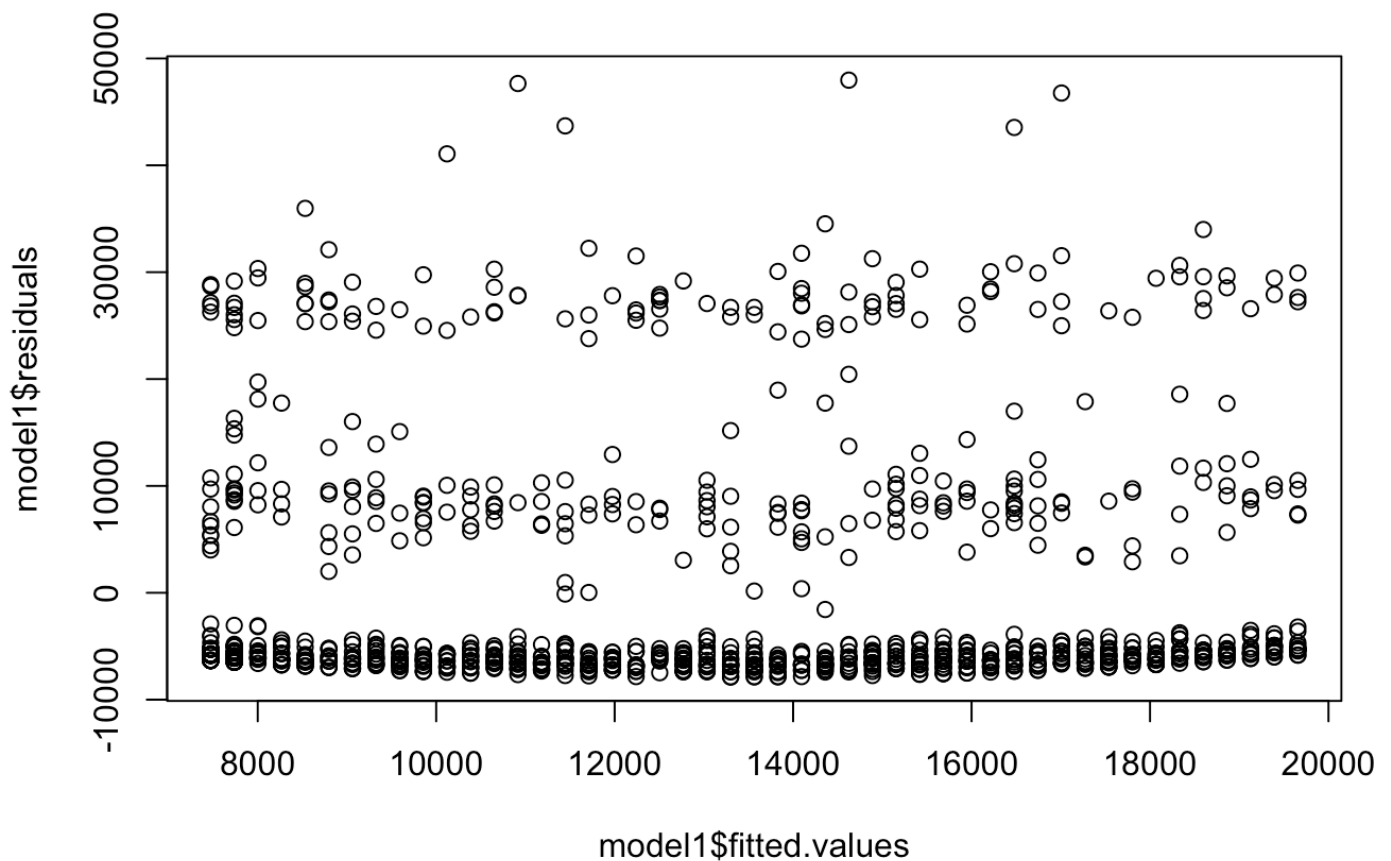
```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:psych':
##
##      logit
```

```
vif(model2)
```

```
##      age      bmi
## 1.011914 1.011914
```

```
plot(model1$fitted.values,model1$residuals) ##funnal test
```

```
library(skedastic)
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
lmtest::bptest(model1)
```

```
##
## studentized Breusch-Pagan test
##
## data: model1
## BP = 0.01957, df = 1, p-value = 0.8887
```

```
lmtest::bptest(model2)
```

```
##
## studentized Breusch-Pagan test
##
## data: model2
## BP = 95.796, df = 2, p-value < 2.2e-16
```

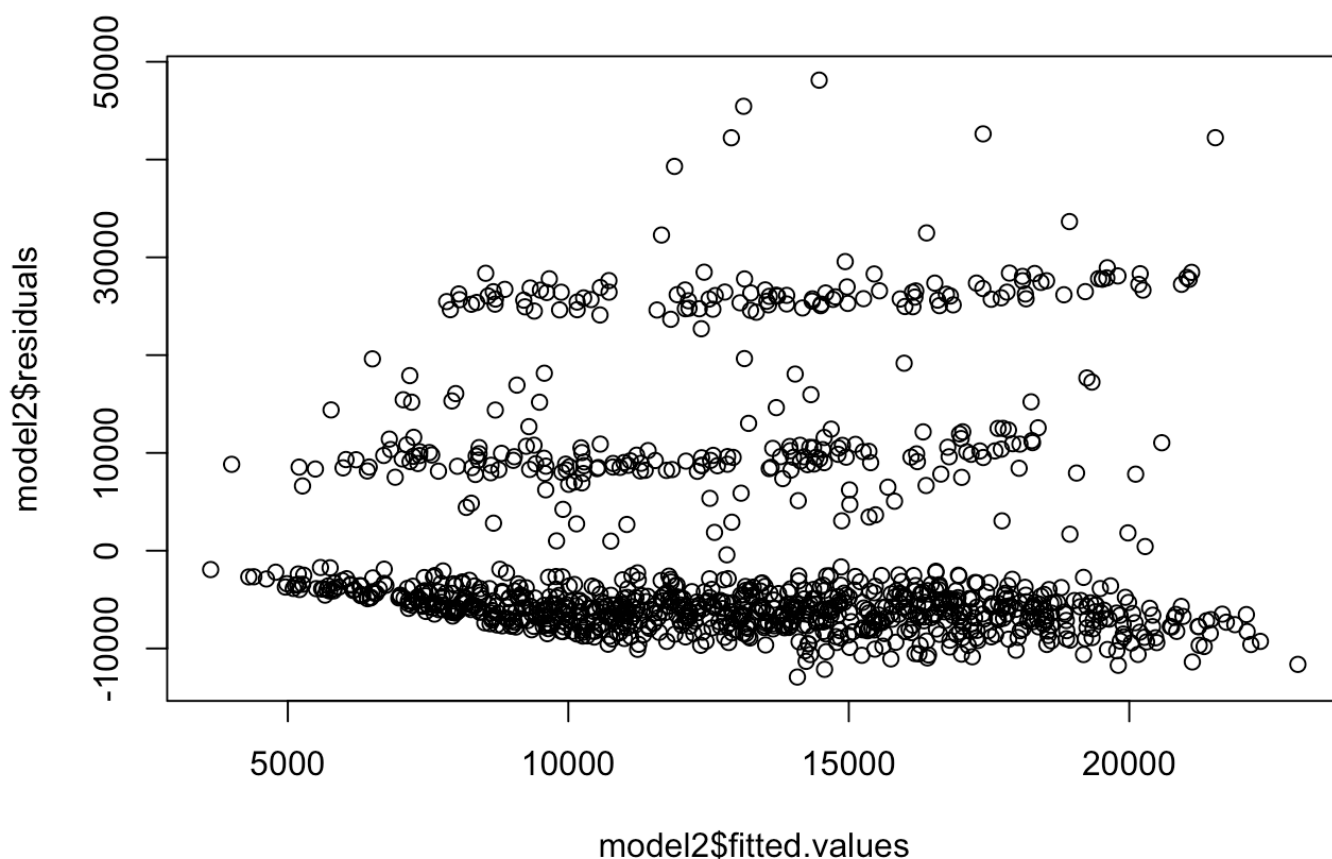
```
skedastic::white(model1)
```

```
## # A tibble: 1 × 5
##   statistic p.value parameter method      alternative
##   <dbl>    <dbl>    <dbl> <chr>      <chr>
## 1      0.583    0.747        2 White's Test greater
```

```
skedastic::white(model2) ##heteroscedasticity -> doesn't want
```

```
## # A tibble: 1 × 5
##   statistic p.value parameter method      alternative
##   <dbl>    <dbl>    <dbl> <chr>      <chr>
## 1      96.9 4.53e-20        4 White's Test greater
```

```
plot(model2$fitted.values,model2$residuals)
```



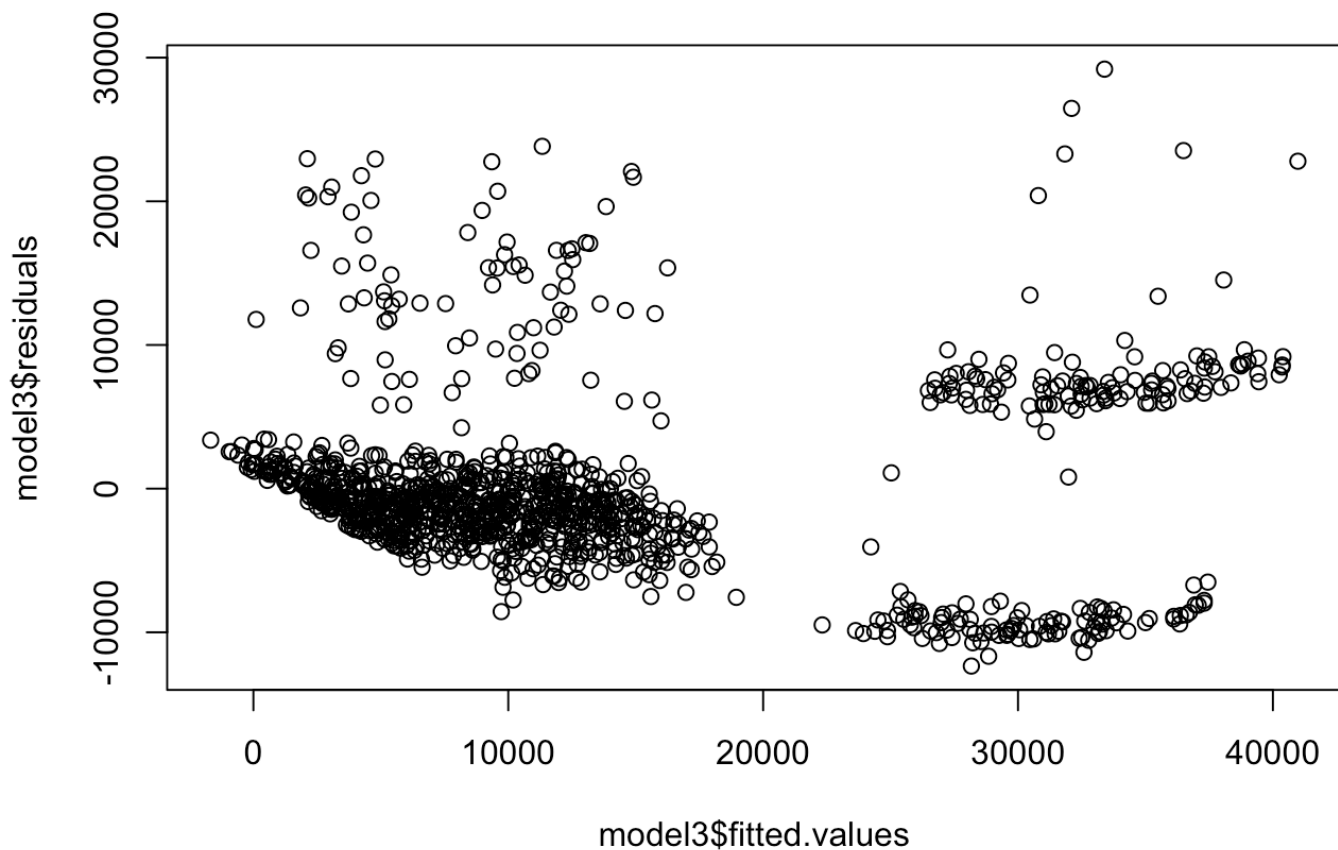
```
###  
model3 = lm(charges ~ age+bmi+smoker, data = training_data)  
summary(model3)
```

```
##  
## Call:  
## lm(formula = charges ~ age + bmi + smoker, data = training_data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -12349  -2952  -1073   1316   29198   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -11285.68    1027.48  -10.98  <2e-16 ***   
## age          261.79      13.04    20.07  <2e-16 ***   
## bmi          306.73      30.15    10.17  <2e-16 ***   
## smoker       23586.97     454.30    51.92  <2e-16 ***   
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 6120 on 1131 degrees of freedom  
## Multiple R-squared:  0.7389, Adjusted R-squared:  0.7382   
## F-statistic: 1067 on 3 and 1131 DF, p-value: < 2.2e-16
```

```
vif(model3)
```

```
##      age      bmi  smoker   
## 1.012140 1.012179 1.000544
```

```
plot(model3$fitted.values,model3$residuals)
```



```
lmtest::bptest(model3)
```

```
##
## studentized Breusch-Pagan test
##
## data: model3
## BP = 101.65, df = 3, p-value < 2.2e-16
```

```
skedastic::white(model3)
```

```
## # A tibble: 1 × 5
##   statistic p.value parameter method      alternative
##   <dbl>     <dbl>     <dbl> <chr>      <chr>
## 1      103. 6.89e-20         6 White's Test greater
```

```
###
```

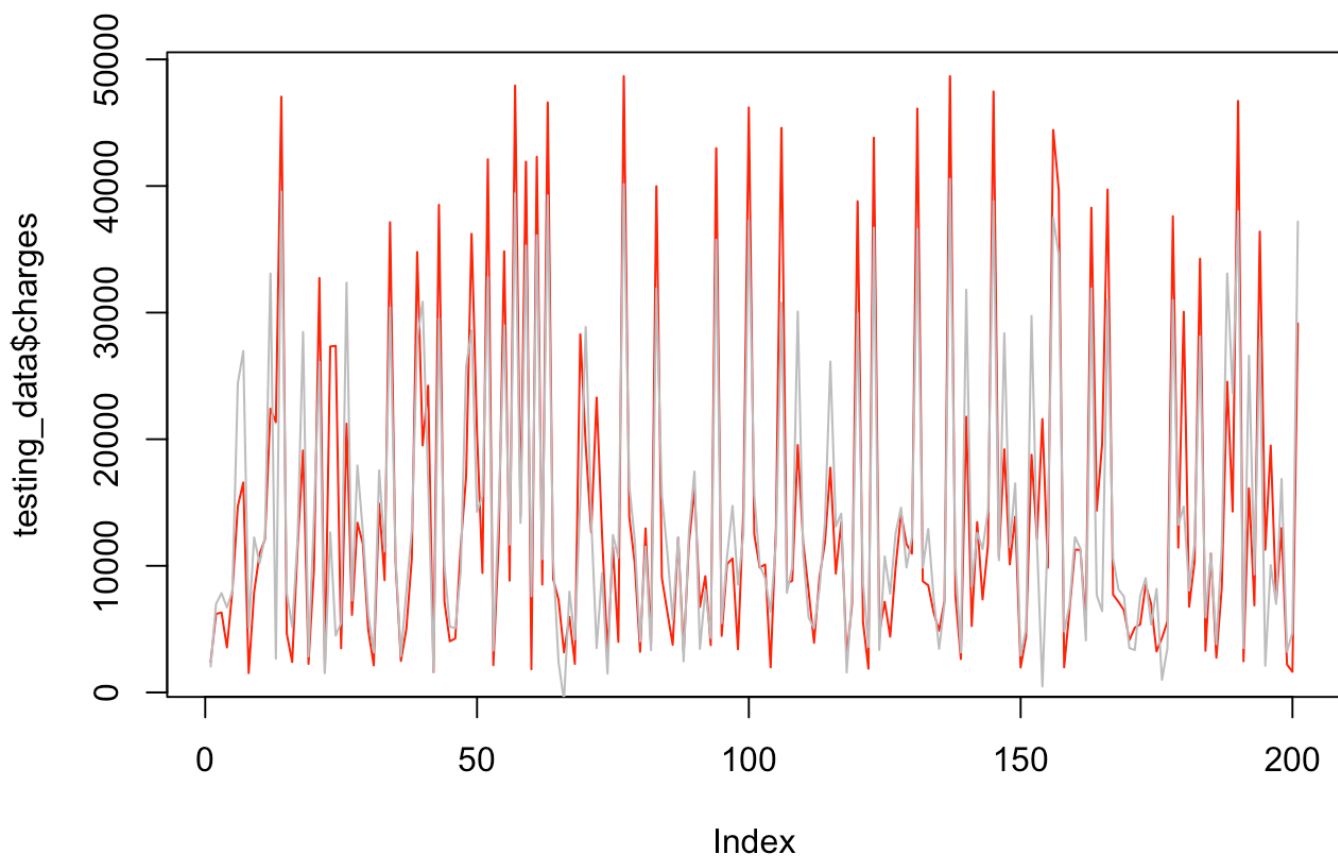
```
## test the model on testing data ##
prediction = predict(model3,testing_data)
head(prediction)
```

```
##          18          25          44          48          55          65
## 2049.637 6996.899 7848.084 6709.669 7986.259 24414.128
```

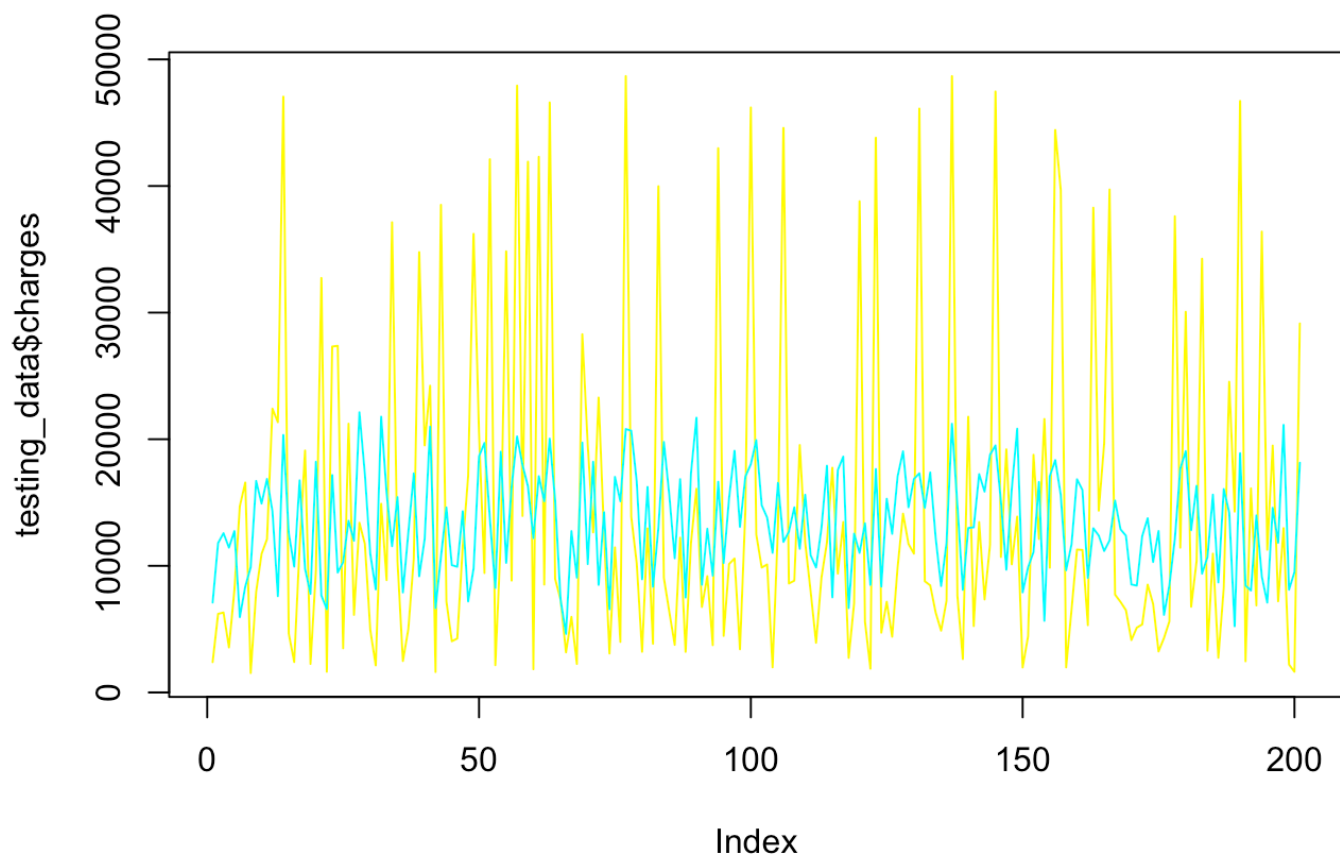
```
head(testing_data$charges)
```

```
## [1] 2395.172 6203.902 6313.759 3556.922 8059.679 14711.744
```

```
plot(testing_data$charges,type="l",col="red")
lines(prediction,type = "l", col="grey")
```



```
prediction1 = predict(model2,testing_data)
plot(testing_data$charges,type="l",col="yellow")
lines(prediction1,type = "l", col="cyan")
```



```
prediction2 = predict(model1,testing_data)
plot(testing_data$charges,type="l",col="yellow")
lines(prediction2,type = "l", col="cyan")
```

